AS/400 → Snowflake Migration Project Plan (12-Week Baseline, Expandable to 6–12 Months)

Phase 1 — Program Initiation & Architecture (Week 1–2)

Objective: Establish governance, architecture, and migration scope.

Key Activities

- **Program charter** creation
- **Snowflake account setup** (roles, warehouses, RBAC)
- **Landing zone provisioning** (S3/ADLS/GCS)
- **Connectivity validation** (ODBC/JDBC, VPN, firewall)
- **Data classification** (PII, PCI, mortgage-sensitive fields)

Deliverables

- Architecture blueprint
- Security & IAM model
- Migration wave plan
- RACI matrix

Phase 2 — AS/400 Discovery & Reverse Engineering (Week 2–4)

Objective: Fully understand legacy data, logic, and dependencies.

Key Activities

- **PF/LF file inventory** (DDS, constraints, keys)
- **RPG III/IV logic analysis** (business rules extraction)
- **Job scheduler mapping** (CL programs, nightly cycles)
- **Data profiling** (packed, zoned, EBCDIC, nullability)

- **CDC feasibility** (journaling, triggers)

Deliverables

- Source-to-target mapping baseline
- RPG logic catalog
- Dependency map
- Data profiling report

Phase 3 — Extraction Framework (Week 4–6)

Objective: Build repeatable extraction pipelines for full + incremental loads.

Key Activities

- **Direct DB2/400 extraction** via ODBC/JDBC
- **Flat-file exports** (fixed-width, EBCDIC → UTF-8)
- **Metadata capture** (row counts, checksums, timestamps)
- **Parallel unloads** for large PFs
- **CDC extraction** using journaling or timestamp logic

Deliverables

- Extraction scripts
- CDC design
- Metadata framework
- Sample raw extracts

Phase 4 — Landing Zone & Snowflake Ingestion (Week 6–8)

Objective: Land AS/400 data in cloud storage and ingest into Snowflake.

Key Activities

- **Raw zone** setup (1:1 source fidelity)

- **Normalized zone** (packed → numeric, EBCDIC → UTF-8)
- **Snowflake stages** (S3/ADLS/GCS)
- **COPY INTO pipelines**
- **Snowpipe** for continuous ingestion
- **Streams & Tasks** for incremental loads

Deliverables

- Landing zone folder structure
- Snowpipe configuration
- COPY scripts
- Ingestion monitoring dashboards

Phase 5 — Transformation (DBT + Snowflake ELT) (Week 8–10)

Objective: Rebuild AS/400 business logic in Snowflake using DBT.

Key Activities

- **DBT staging models** (renaming, casting, cleaning)
- **DBT intermediate models** (joins, logic reconstruction)
- **DBT marts** (analytics-ready structures)
- Convert RPG logic → SQL/DBT
- Implement SCD Type 1/2
- Apply clustering keys & micro-partition optimization

Deliverables

- DBT project
- Lineage graph
- Transformation specifications
- Performance tuning report

Phase 6 — Validation & Reconciliation (Week 10–11)

Objective: Ensure accuracy, completeness, and audit readiness.

Key Activities

- **Schema validation**
- **Metric validation** (row counts, sums, distincts)
- **Row-level validation** for mortgage/financial data
- **Functional validation** (legacy vs Snowflake outputs)
- Hash-based reconciliation
- DBT tests

Deliverables

- Validation reports
- Reconciliation dashboards
- Audit logs

Phase 7 — Cutover & Go-Live (Week 11–12)

Objective: Switch from AS/400 to Snowflake with minimal disruption.

Key Activities

- Freeze window
- Final CDC replay
- Parallel run (legacy + Snowflake)
- Business sign-off
- Downstream system cutover
- Rollback plan validation

Deliverables

- Cutover runbook
- Go-live checklist

- Communication plan

Phase 8 — Decommissioning & Optimization (Post Go-Live)

Objective: Retire legacy systems and optimize Snowflake for long-term use.

Key Activities

- Archive AS/400 data
- Retire PF/LF files, RPG jobs, CL programs
- Shut down LPARs
- Snowflake cost optimization
- RBAC/ABAC governance hardening

Deliverables

- Decommission checklist
- Cost optimization plan
- Final architecture documentation

High-Level Timeline (12-Week Baseline)

Phase	Duration	Output
Initiation	Week 1–2	Architecture, governance
Discovery	Week 2–4	Inventory, profiling, mappings
Extraction	Week 4–6	Extract scripts, CDC
Landing + Ingestion	Week 6–8	Raw → Snowflake ingestion
Transformation	Week 8–10	DBT models, marts
Validation	Week 10–11	Reconciliation

Phase	Duration	Output
Cutover	Week 11–12	Go-live
Optimization	Post-Go-Live	Cost + performance tuning

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